In the next chapters, we will see how these requirements can be translated into mathematical expressions to model a *cost function* that can reflect how good a graph is.

We'll use the rest of this chapter, instead, to talk more diffusely about the third point.

As we have seen, there are graphs for which it's not possible to find an embedding without any edge crossing.

Interesting enough, if we move from the plane to 3D space, it becomes trivial to find an embedding to $\mathbb{R}^3$ such that no edges cross: if we consider the following Jordan curve `C(t)`, a surface defined as

$$C(t) = \begin{cases} x = t \\ y = t^2 \\ z = t^3 \end{cases}, t \geq 0$$

then we can map each vertex to a distinct point on the curve, and draw edges as segments between the vertices. It can then be proven that there is no way to choose a set of 4 points from `C(t)` such that they all lie on the same plane, and therefore the segments between pairs of points can't cross each other.

For non-planar graphs `G`, however, we can define a value, called **crossing number**, that is the smallest possible number of edge crossings of all the possible embeddings of `G` to $\mathbb{R}^2$.